

FCC 3101 CHAPTER 3

**AREAS OF
SPECIALIZATION**

PART 2

3.5 WATER RESOURCES ENGINEERING

3.5.1 Area of Specialization: Water Resources Engineering specializes in the design, construction and maintenance of:

1. Water supply systems
2. Hydroelectric power stations
3. Groundwater and rainwater harvesting systems
4. Irrigation systems
5. Storm drainage, flood mitigation and control systems

3.5.2 Core Course Units: Water Resources Engineering is covered by the following core course units: -

- 1. Fluid Mechanics** FCC 3222, FCE 322, FCE 421, FCE 422
- 2. Hydrology:** FCE 425
- 3. Water Resources:** FCE 525, FCE 526

3.5.3 Common Course Units [Same as 3.2.3]

3.5.4 Support Course Units [Same as 3.2.4]

3.5.5 Foundation Course Units [Same as 3.2.5]



Fig 3.11 Masonry Dam



Fig 3.12 RC Dam (Showing Spillway)

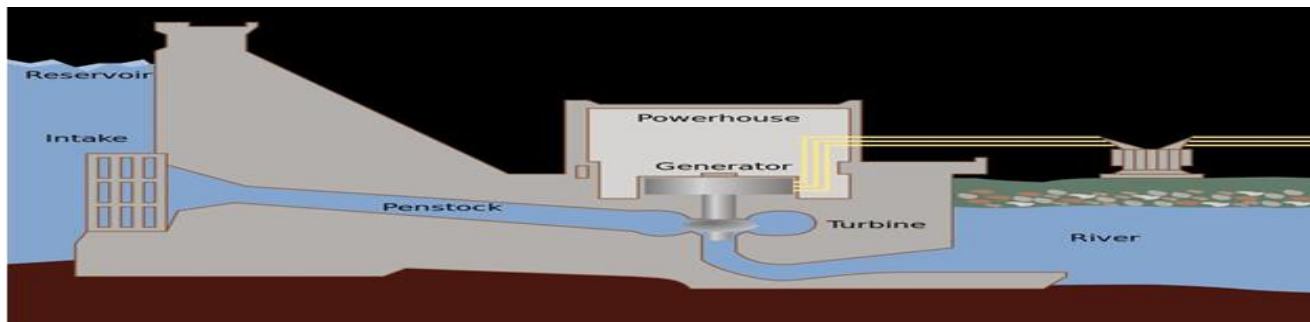


Fig 3.13 Schematic Drawing of a Hydroelectric Power Station

3.6 PUBLIC HEALTH ENGINEERING

3.6.1 Area of Specialization: Public Health Engineering (aka **Environmental Health Engineering**) specializes in the design, construction and maintenance of:

1. Waste water collection, treatment, recycling and safe disposal systems
2. Solid waste collection, treatment, recycling and safe disposal systems
3. Industrial waste water treatment and disposal systems

3.6.2 Core Course Units: Public Health Engineering is covered by the following core course units: -

1. **Water and wastewater treatment** FCE 481, FCE 482
2. **Water and wastewater engineering design** FCE 581
3. **Industrial waste treatment:** FCE 582

3.6.3 Common Course Units [Same as 3.2.3]

3.6.4 Support Course Units [Same as 3.2.4 + **FCE 391**]

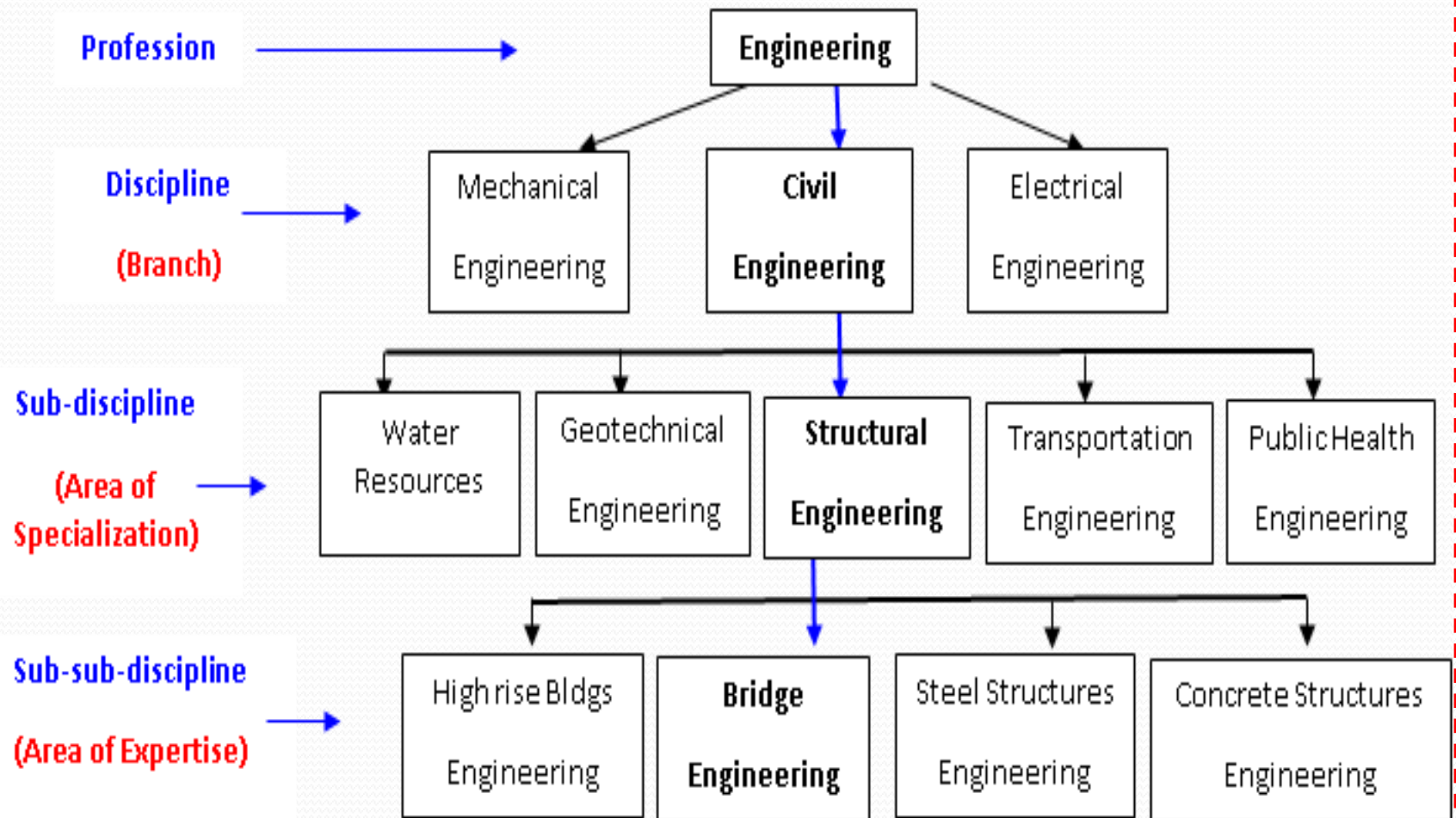
3.6.5 Foundation Course Units [Same as 3.2.5]



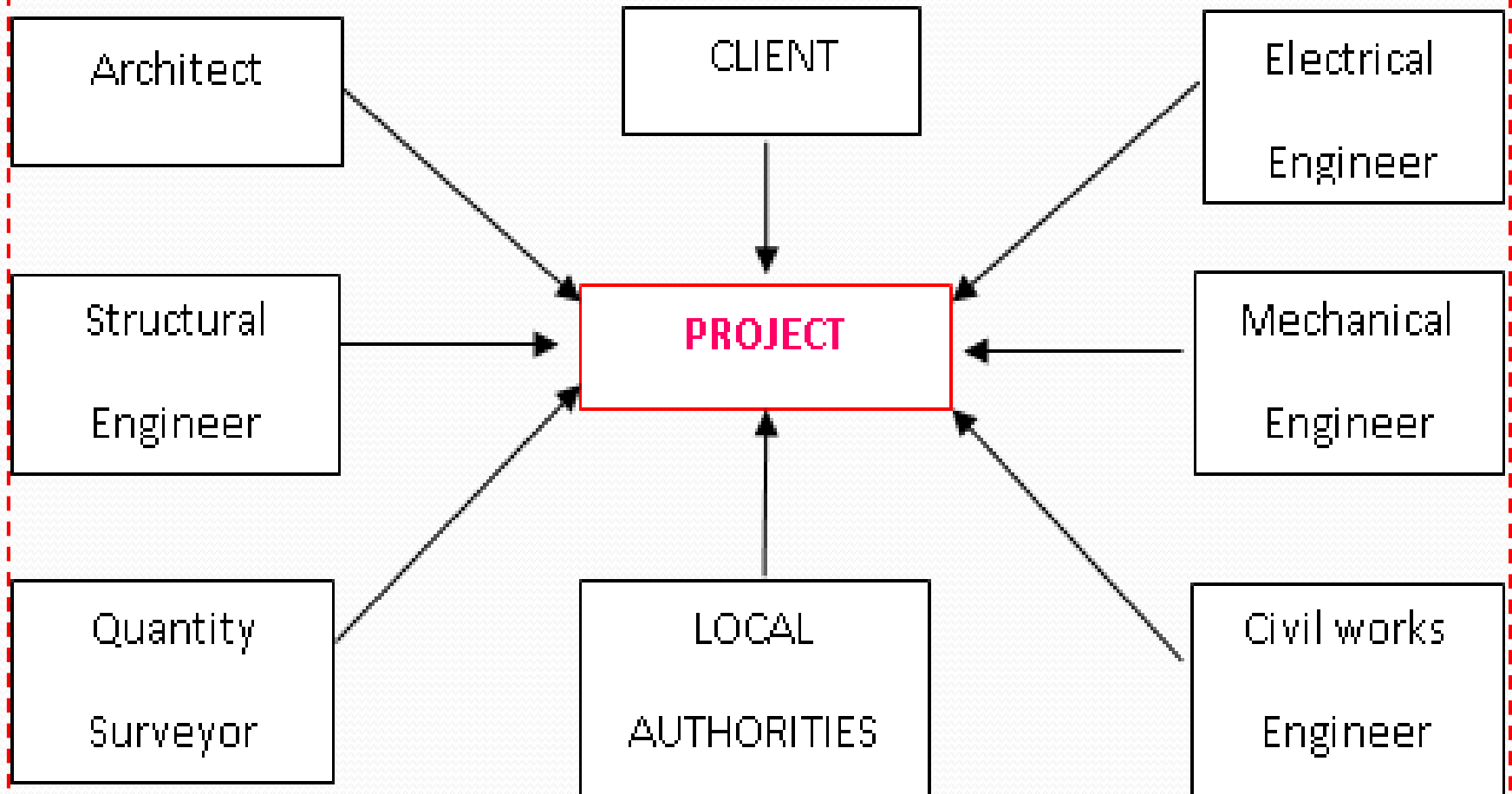
Fig 3.14 Wetlands Sewage Treatment Plant

ADDITIONAL INFORMATION

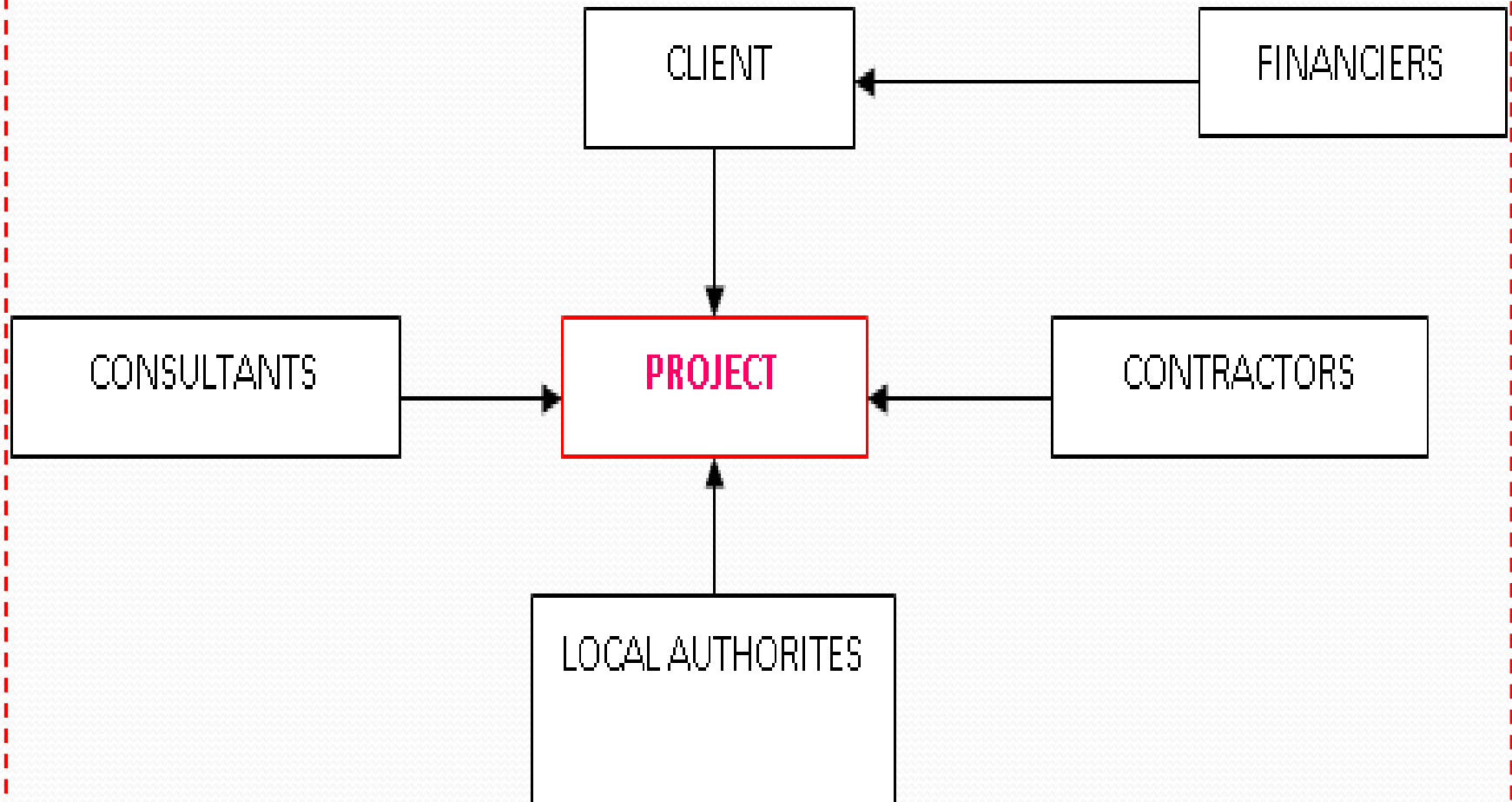
3.7 ORGANIZATIONAL CHART



3.8 PROJECT DESIGN



3.9 PROJECT IMPLEMENTATION



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